Technical University of Crete

School of Electrical and Computer Engineering

Course: Reinforcement Learning and Dynamic Optimization

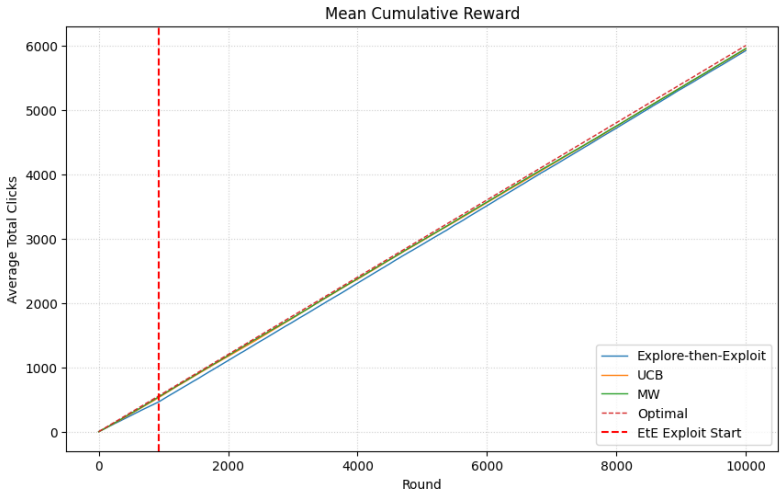
Programming Assignment 2 (First part)

Algorithms to implement

For **Explore-then-Exploit (EtE)**, we set the exploration phase length to $N = T^{2/3}$, alternating arms before committing to the empirical best. For **UCB**, we use the confidence bound $\sqrt{2\log(T)/N_i(t)}$, pulling each arm once initially. Finally, for **Multiplicative Weights (MW)**, we choose the exploration rate ϵ and the learning rate η to be equal, specifically $\epsilon = \eta = \sqrt[3]{\frac{K \ln K}{T}}$.

Sanity Check (Stationary)

The environment is made stationary by setting $p_i^L = p_i^H$ for each arm, so that $p_i(t)$ remains constant over time. In particular, we set $p_0^L = p_0^H = 0.4$ and $p_1^L = p_1^H = 0.6$.



The results confirm that all algorithms behave sensibly, with cumulative reward converging to the optimal expected reward.

Algorithm	**Total Clicks**
EtE	5920.80 ± 42.59
UCB	5946.10 ± 56.54
MW	5953.50 ± 64.87
Optimal Expected	6000.00

Environment where MW beats EtE

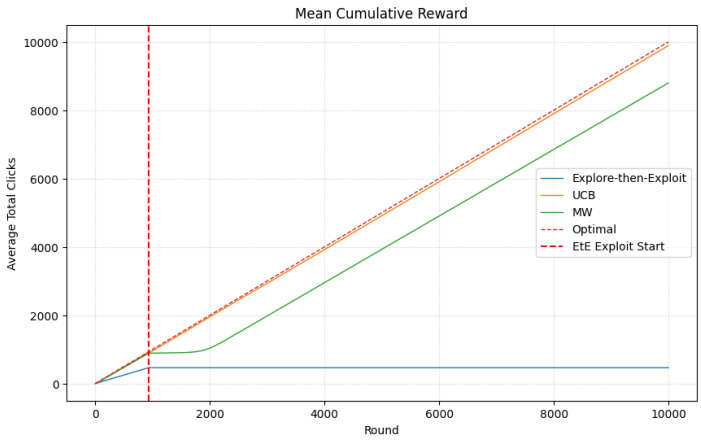
To cause Explore-then-Exploit (EtE) to fail, we construct an environment where Arm 1 appears optimal during the exploration phase ($t \leq 2N$), but its reward drops to 0 exactly when exploitation begins. As parameterized below, EtE falsely commits to Arm 1 and collects zero subsequent reward.

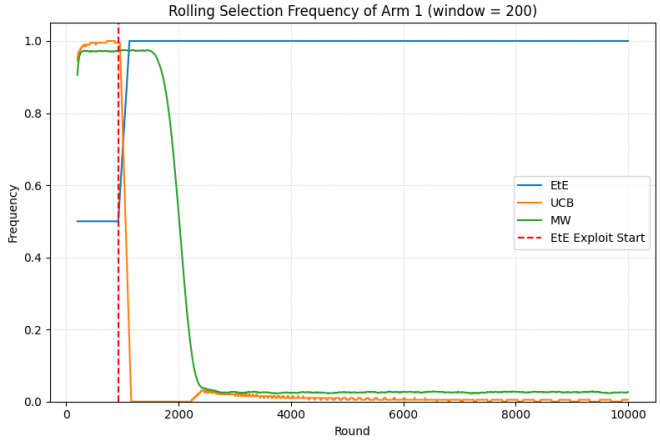
Parameter	**Arm 0**	**Arm 1**
p^L	0	0
p^H	1	1
T^L	$2N$	$T - 2N$
T^H	$T - 2N$	$2N$
Start	Low	High

The expected cumulative reward for EtE for this allocation is given by:

$$\begin{aligned}\mathbb{E}[R_{\text{EtE}}] &= \mathbb{E}[R_{\text{explore},0}] + \mathbb{E}[R_{\text{explore},1}] + \mathbb{E}[R_{\text{exploit},1}] \\ &= N \cdot p_0^L + N \cdot p_1^H + (T - 2N) \cdot p_1^L \\ &= N \cdot 0 + N \cdot 1 + (T - 2N) \cdot 0 \\ &= N\end{aligned}$$

Our simulations confirm this theoretical failure, illustrating both the flatlining cumulative reward and the arm selection behavior that causes it. While MW experiences a temporary dip in performance following the environment shift at round $2N$, it successfully adapts, easily beating the stalled EtE algorithm.





Algorithm	Total Clicks
EtE	464.00 ± 0.00
UCB	9892.00 ± 0.00
MW	8798.40 ± 146.12
Optimal Expected	10000.00

Environment where MW beats UCB

To induce underperformance in UCB, we exploit its reliance on historical averages, which makes it vulnerable in non-stationary environments where the optimal arm changes over time.

Parameter	Arm 0	Arm 1
p^L	0	0.45
p^H	1	0.9
T^L	$0.575 \cdot T$	$0.575 \cdot T$
T^H	$0.425 \cdot T$	$0.425 \cdot T$
Start	High	High

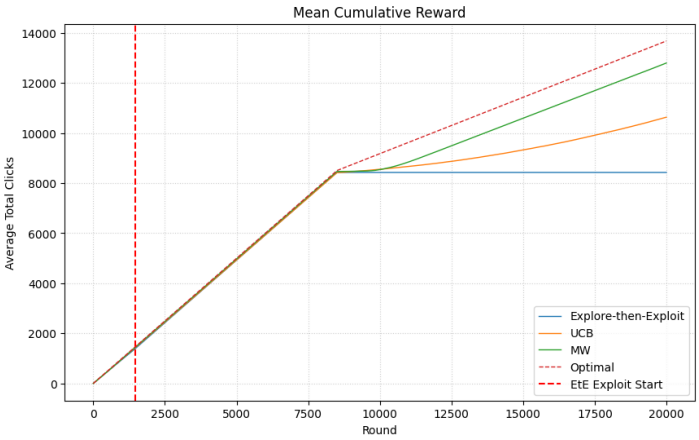
During the first phase, we make one arm clearly superior so that UCB selects it repeatedly and builds a strong empirical mean. We then switch the phases late enough for this estimate to be well established, while still leaving enough time for UCB's slow adaptation to become visible.

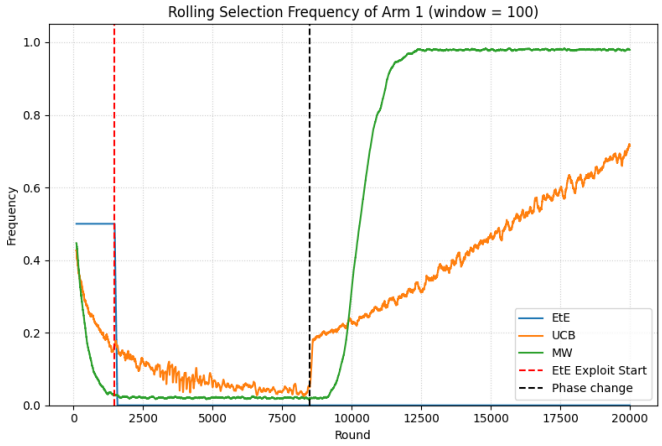
As a result, the initially optimal arm is selected for about 90% of the first phase on average, giving it an empirical mean of 1. After the switch, this estimate decreases very slowly, even though the arm starts producing zero rewards, so UCB keeps selecting it for a long time and therefore performs worse than MW.

Regarding the choice of probabilities, the reward probability of the old arm in the Low phase is set to 0 so that it becomes truly suboptimal. At the same time, the new optimal arm is given a higher, but still moderate, reward probability. This makes it truly better, but not immediately obvious to UCB. Since this arm remains noisy, post-switch exploration does not provide a strong enough evidence for fast adaptation.

Algorithm	Total Clicks
UCB	10632.54 ± 117.14
MW	12794.84 ± 261.20

Metric (MW vs UCB)	Value
Average Click	2162.30 ± 283.85
Percentage	$20.35\% \pm 2.77\%$





As can also be seen from the rolling frequency plot of the arm that becomes optimal after the switch, UCB essentially never learns to consistently play it, or more accurately, it does not have enough time to do so, since the selection frequency of the optimal arm increases only gradually and never reaches 100% by the end of the horizon.

However, by keeping the duration of the initial phase constant at 8500 rounds but extending the experiment until UCB eventually converges on the new optimal arm, we determine that UCB requires a total horizon of 45000 rounds. As expected over this longer timeframe, the total performance improvement is reduced, resulting in a improvement of $14.52\% \pm 1.78\%$.

